

Nu(Pr , Ra) Dataset: Sources and Provenance

Machine-learning prediction of the Nusselt number in 2D Rayleigh-Bénard convection

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Overview

This document records exactly where every number in `data/nu_dataset_model.csv` came from. Section 3.3 of Shreshthi & Pandey’s paper (*Turbulent Prandtl number in the near-wall region of thermal convection*) trains a random forest regressor to predict the Nusselt number Nu from the Prandtl number Pr and Rayleigh number Ra , using 55 data points drawn from five cited papers plus the paper’s own simulations. This document reproduces the relevant tables from the four sources we have obtained so far, highlights which rows were kept, and comments on why the rest were left out.

 **Included** in `nu_dataset_model.csv`  **Excluded** (see Comment column)

References

1. Shreshthi and A. Pandey. *Turbulent Prandtl number in the near-wall region of thermal convection*. Project paper (`papers/shreshthi_pandey_2026_turbulent_prandtl_number.pdf`).
2. A. Pandey. *Thermal boundary layer structure in low-Prandtl-number turbulent convection*. J. Fluid Mech., 910:A13, 2021. [doi:10.1017/jfm.2020.961](https://doi.org/10.1017/jfm.2020.961) — [arXiv:2101.03051](https://arxiv.org/abs/2101.03051).
3. A. Pandey and K. R. Sreenivasan. *Transient and steady convection in two dimensions*. J. Fluid Mech., 1015:A42, 2025. [doi:10.1017/jfm.2025.10357](https://doi.org/10.1017/jfm.2025.10357) — [arXiv:2503.03080](https://arxiv.org/abs/2503.03080).
4. A. Pandey, H. Tiwari, and K. R. Sreenivasan. *Thermal convection in 1, 2, 3, and 4 dimensions*. J. Fluid Mech., in press, 2026. [arXiv:2607.26372](https://arxiv.org/abs/2607.26372).
5. E. P. van der Poel, R. J. A. M. Stevens, and D. Lohse. *Comparison between two- and three-dimensional Rayleigh-Bénard convection*. J. Fluid Mech., 736:177–194, 2013. [doi:10.1017/jfm.2013.488](https://doi.org/10.1017/jfm.2013.488). **Not yet obtained** — reports Nu only as scatter plots (Fig. 5a, 6b), not a table; paywalled, needs institutional access to digitize.
6. Y. Zhang and Q. Zhou. *Low-Prandtl-number effects on global and local statistics in two-dimensional Rayleigh-Bénard convection*. Phys. Fluids, 36(1):015107, 2024. [doi:10.1063/5.0175011](https://doi.org/10.1063/5.0175011). **Not yet obtained** — fully paywalled, no open copy found; needs institutional access.

All four obtained sources describe the *same* physical setup: a closed 2D square box ($\Gamma = 1$), no-slip walls, isothermal top/bottom plates, adiabatic sidewalls, solved with the same spectral-element code. This is confirmed empirically, not just from the papers’ text: several rows below report the exact same Nu to the decimal at the same Pr, Ra in two different papers — only possible if it is literally the same simulation, reused across papers.

1 Present paper — Shreshthi & Pandey, Table 1

Closed 2D box, $Ra = 10^9$ fixed, Pr varied. All 13 rows are the paper’s own flagship simulations and are included.

Table 1: Present paper, Table 1 ($Ra = 10^9$).

Pr	Ra	Nu	Comment
0.021	10^9	37.43	Included
0.05	10^9	41.51	Included
0.1	10^9	42.72	Included
0.2	10^9	45.98	Included
0.35	10^9	48.1	Included
0.5	10^9	48.64	Included
0.7	10^9	49.22	Included
1	10^9	50.87	Included
2	10^9	51.09	Included
4.38	10^9	52.67	Included
7	10^9	52.27	Included
20	10^9	53.41	Included
50	10^9	52.12	Included

2 Pandey (2021), Table 1

Closed 2D box, $Pr = 0.021$ fixed, Ra varied. Runs labelled with an ‘a’ (e.g. 3a) are resolution checks of the preceding run — the same physical point, re-run at finer mesh, to confirm numerical convergence. Only the base run is kept; its resolution-check twin is excluded since it is not an independent data point.

Table 2: Pandey (2021), Table 1 ($Pr = 0.021$).

Run	Ra	Nu	Comment
1	5×10^5	5.2	Included
1a	5×10^5	5.2	Resolution check of run 1
2	10^6	6.3	Included
2a	10^6	6.3	Resolution check of run 2
3	10^7	10.8	Included
3a	10^7	11.1	Resolution check of run 3
4	10^8	19.6	Included
5	10^9	35.4	Included
5a	10^9	36.2	Resolution check of run 5

3 Pandey, Tiwari & Sreenivasan (2026, in press)

Table 1 — closed 2D box, $Pr = 0.021$ and $Pr = 0.05$

The $Ra = 10^9$ rows for both Pr values report exactly the same Nu as the present paper’s own Table 1 (37.43 and 41.51 respectively) — the same simulation, reused across papers. Excluded here to avoid

counting it twice.

Table 3: Pandey, Tiwari & Sreenivasan (2026), Table 1.

Pr	Ra	Nu	Comment
0.021	10^7	10.97	Included
0.021	2×10^7	13.04	Included
0.021	3×10^7	14.42	Included
0.021	5×10^7	16.58	Included
0.021	10^8	19.68	Included
0.021	4×10^8	28.32	Included
0.021	10^9	37.43	Matches Present paper exactly – duplicate
0.05	10^7	12.41	Included
0.05	2×10^7	14.52	Included
0.05	3×10^7	16.06	Included
0.05	10^8	21.64	Included
0.05	3×10^8	29.27	Included
0.05	5×10^8	33.5	Included
0.05	10^9	41.51	Matches Present paper exactly – duplicate

Table 2 — “RBC-P”, $Pr = 1$, $D = 2$ column

This table uses a different solver (DHARA, finite-difference, versus the spectral-element code used everywhere else) for a 1D/2D/3D/4D dimension-comparison study. It is a *different physical configuration* from the closed box used in every other table here: at $Pr = 1, Ra = 10^8$ it reports $Nu = 13.6$, versus $Nu = 25.01$ in Pandey & Sreenivasan (2025)’s Table 1 at the identical Pr, Ra (Table 5). That gap is far too large to be numerical noise. **All rows here are excluded** from the training set.

Table 4: Pandey, Tiwari & Sreenivasan (2026), Table 2 (RBC-P, $D = 2$, $Pr = 1$).

Ra	Nu	Comment
2×10^5	4.8	Different configuration (RBC-P) – excluded
5×10^5	6.2	Different configuration (RBC-P) – excluded
10^6	6.9	Different configuration (RBC-P) – excluded
2×10^6	8.5	Different configuration (RBC-P) – excluded
5×10^6	10.3	Different configuration (RBC-P) – excluded
10^7	11.8	Different configuration (RBC-P) – excluded
2×10^7	13.1	Different configuration (RBC-P) – excluded
5×10^7	13.3	Different configuration (RBC-P) – excluded
10^8	13.6	Different configuration (RBC-P) – excluded
2×10^8	15.5	Different configuration (RBC-P) – excluded
5×10^8	20.4	Different configuration (RBC-P) – excluded
10^9	24.6	Different configuration (RBC-P) – excluded

4 Pandey & Sreenivasan (2025), Table 1

Closed 2D box, $Pr = 0.1$ and $Pr = 1$, wide Ra range. As with the 2026 paper, the $Ra = 10^9$ rows duplicate the present paper’s own simulations exactly (42.72 and 50.87) and are excluded.

Table 5: Pandey & Sreenivasan (2025), Table 1.

Pr	Ra	Nu	Comment
0.1	10^6	5.91	Included
0.1	2×10^6	7.05	Included
0.1	3×10^6	7.88	Included
0.1	6×10^6	10.04	Included
0.1	10^7	11.77	Included
0.1	2×10^7	15.35	Included
0.1	3×10^7	17.5	Included
0.1	10^8	23.01	Included
0.1	3×10^8	31.06	Included
0.1	10^9	42.72	Matches Present paper exactly – duplicate
0.1	3×10^9	59.15	Included
0.1	10^{10}	84.74	Included
1	3×10^6	6.01	Included
1	6×10^6	7.11	Included
1	8×10^6	8.18	Included
1	9×10^6	8.64	Included
1	10^7	9.03	Included
1	1.5×10^7	13.33	Included
1	2×10^7	14.75	Included
1	3×10^7	17.27	Included
1	6×10^7	21.48	Included
1	10^8	25.01	Included
1	3×10^8	35.61	Included
1	6×10^8	44.15	Included
1	10^9	50.87	Matches Present paper exactly – duplicate
1	3×10^9	67.51	Included
1	10^{10}	94.52	Included
1	3×10^{10}	130.2	Included
1	10^{11}	183.3	Included
1	3×10^{11}	262.3	Included
1	10^{12}	394.2	Included

5 Summary

Table 6: Row counts by source.

Source	Rows in table	Included	Excluded
Present paper, Table 1	13	13	0
Pandey (2021), Table 1	9	5	4
Pandey, Tiwari & Sreenivasan (2026), Table 1	14	12	2
Pandey, Tiwari & Sreenivasan (2026), Table 2 (RBC-P)	12	0	12
Pandey & Sreenivasan (2025), Table 1	31	29	2
Total	79	59	20

The 59 included rows across all four sources make up `data/nu_dataset_model.csv`. Once the two missing papers (references 5 and 6) are obtained, their tables should be added here following the same process: reproduce the table, confirm the physical setup matches (closed box, same boundary conditions) before including any row, and update the summary above.